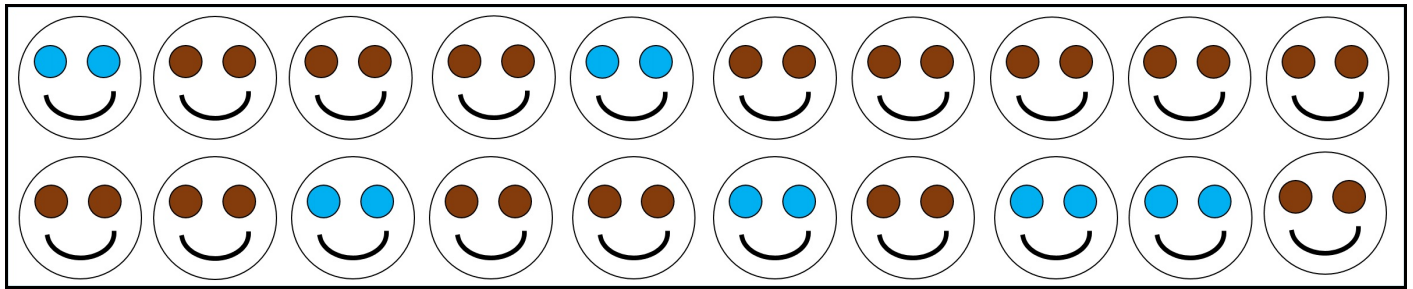


PROBLEM #1.

You have sampled a population for their eye color and only two phenotypes exist, BROWN (which is dominant) and BLUE (which is recessive). If the brown allele ("A") is dominant over the recessive allele ("a") and these are the ONLY two alleles in the population, answer the following questions. The data set for the population is shown below. Your answers should be to one decimal place and expressed as a percentage (i.e. 0.4768 would be 47.7%)



- A. The frequency of the recessive phenotype.
- B. The frequency of the dominant phenotype.
- C. The frequency of the "a" allele.
- D. The frequency of the "A" allele.
- E. The frequency of the people who have BROWN eyes that are homozygous for their genotype.
- F. The frequency of the people who have BROWN eyes that are heterozygous for their genotype.

PROBLEM #2.

Sickle-cell anemia is an interesting genetic disease. Normal homozygous individuals (SS) have normal blood cells that are easily infected with the malarial parasite. Thus, many of these individuals become very ill from the parasite and many die. Individuals homozygous for the sickle-cell trait (ss) have red blood cells that readily collapse when deoxygenated. Although malaria cannot grow in these red blood cells, individuals often die because of the genetic defect. However, individuals with the heterozygous condition (Ss) have some sickling of red blood cells, but generally not enough to cause mortality. In addition, malaria cannot survive well within these "partially defective" red blood cells. Thus, heterozygotes tend to survive better than either of the homozygous conditions.

If 9% of an African population is born with a severe form of sickle-cell anemia (ss), then answer these questions.

- A. What percentage of the population will be more resistant to malaria because they are heterozygous (Ss) for the sickle-cell gene?

PROBLEM #3.

Within a population of butterflies, the color brown (B) is dominant over the color white (b). And, 40% of all butterflies are white. Given this simple information, which is something that is very likely to be on an exam, calculate the following:

- A. The percentage of butterflies in the population that are heterozygous.
- B. The frequency of homozygous dominant individuals.

PROBLEM #4.

A rather large population of organisms have 396 red-sided individuals and 557 tan-sided individuals. Assume that red is totally recessive. Please calculate the following:

- A. The allele frequencies of each allele.
- B. The expected genotype frequencies.
- C. The number of heterozygous individuals that you would predict to be in this population.
- D. The expected phenotype frequencies.
- E. Conditions happen to be really good this year for breeding and next year there are 1,245 offspring. Assuming that all of the Hardy-Weinberg conditions are met, how many of these would you expect to be red-sided and how many tan-sided?